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Stereo Camera DID Specification

V1.2

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0. Revision History

Date	Version	Modification	Notes
2021.01.07	V1.0	Initial version	Written by: ZhouDingduo Revised by: Liwenfeng, HuangZhi
2021.05.31	V1.1	Add description about signal arrangement	Written by: ZhouDingduo Revised by: Liwenfeng, HuangZhi
2022.03.30	V1.2	Add F200 dynamic calibration progress	Written by: WenNing Revised by: ZhouDingduo

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1. DID Specification

DID (Hex)	Description	Length [bytes]	Unit	Value	Boot Server		Application Server		Comments
					Read	Write	Read	Write	
F187	Vehicle manufacturer spare part number	7	ASCII	XXXXXXX			X	X	Value to be defined by customer.
F188	vehicleManufacturerECUSoftwareNumber	12	ASCII	XXXXXXXXXXXX			X	X	Value to be defined by customer.
F189	vehicleManufacturerECUSoftwareVersionNumber	8	ASCII	XXXXXXXX			X	X	Value to be defined by customer.
F18A	systemSupplierIdentifier	12	ASCII	“Knorr-Bremse”			X		See UDS spec for further details
F18B	ECUManufacturingDateandSerialNumber	9	ASCII	“YYWWDXXXX”			X		Set by during BLFLC25 assembly, XXXX is serial number of ECU
F190	VIN	17	ASCII	INIT value is 00h			X	X	Optional value. Can be set by vehicle manufacturer at EOL process
F191	vehicleManufacturerECUHardwareNumber	15	ASCII	XXXXXXXXXXXXXXX			X	X	Value to be defined by customer.
F192	systemSupplierECUHardwareNumber	11	ASCII	XXXXXXXXXXX			X		Set in ECU production
F194	systemSupplierECUSoftwareNumber	8	ASCII	XXXXXXXX			X		Constant in application software.
F197	systemNameOrEngineType	7	ASCII	"BLFLC25"			X		Constant in application software

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Signal Type	DID (Hex)	Description	Length [bytes]	Data Bytes	Unit	Type	Conversion	Value	Boot Server		Application Server		Comments
									Read	Write	Read	Write	
Vehicle Calibration - ECU Configuration	FD10	Left Wheel	2	B4-B5	cm	u16	1 cm/ bit	Min 20, Max 200	x		x	x	
		Right Wheel	2	B6-B7	cm	u16	1 cm/ bit	Min 20, Max 200	x		x	x	The camera checks the validation of: **1. Left Wheel 2. Right Wheel 3. Left + Right Wheel ≤ 4 meters**
		Distance To Head	2	B8-B9	cm	u16	1 cm/ bit	Min 0, Max 255	x		x	x	Longitudinal coordinate of reference point. (i.e., longitudinal distance from camera to front bumper).
		Lateral to Center Line	2	B10-B11	cm	s16	1 cm/ bit	Min -100, Max 100	x		x	x	Lateral coordinate of reference point. (i.e., lateral distance from camera to middle of front bumper. **The camera doesn't check the validation.**
		Chassis Number	4	B12-B15	int	u32	None	Min 0, Max 4294967296	x		x	x	Optional to be written by vehicle manufacturer. Value not used by SW, but has to be in range
		Country Code	1	B16	enum	u8	None	1 = China	x		x	x	Respective country of main vehicle usage must be written by vehicle manufacturer
		Status	1	B17		u8	None	0x55 - Not Programmed 0xAA - Programmed	x		x	x	Must be set to 0xAA during calibration

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Base Camera Calibration	FD11	FoeX (yaw)	2	B4-B5	Pixels	s16		Min -150 Max 150	x		x		Values after calibration. Only for debugging purposes. If FoeX out of valid range, check: **The camera and SPTAC is on the symmetry line.**
		FoeY (horizon)	2	B6-B7	Pixels	s16		Min -240 Max 120	x		x		If FoeY out of valid range, check: **1. The camera is installed firmly 2. The SPTAC height to floor should be 90cm**
		Roll Angle	2	B8-B9	rad	s16	(value/8192)=radians	Min -450 Max 450	x		x		If Roll Angle out of valid range, check: **1. The camera is installed firmly 2. The SPTAC has no horizontal rotation**
		Camera Height	2	B10-B11	cm	u16	1 cm/ bit	Min 50, Max 300	x		x		If Camere Height out of valid range, check: **1. The camera install height 2.The SPTAC height to floor should be 90cm**
		Grabbing Shift	1	B12	Pixels	u8		Min 0, Max 54	x		x		
		Status	1	B13		u8	None	0x55 - Not Calibrated 0xAA - Calibrated	x		x		Value must be equal to "0xAA" after sucessfull SPTAC calibration..
Dynamic Calibration Progress	F200	Range of dynamic calibration	1	B4	%	u8	1%	Min 0, Max 100 PH=INT {0,100} % [0x0,0x64]	x		x		Percentage of dynamic calibration progress

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		DC State	1	B5		u8	None	DC State [] = [0x00-Undefined 0x01-Start 0x02-In Progress 0x03-Finish (successful) 0x04-Terminate (Error) 0x05-Not In Progress (PIs trigger dynamic calibration)]	x		x		0x00-Undefined: Not used. 0x01-Start : When Camera received dynamic calibration command. 0x02-In Progress: Dynamic Calibration Started, not finished. 0x03-Finish: Dynamic Calibration finished and successful. 0x04-Terminate: Dynamic Calibration Failed. 0x05-Not InProgress: Dynamin Calibration not started.

Note : Signals in DID FD10 and FD11 are arranged with MOTOROLA format. For example, high byte of LeftWheel in DID FD10 is B4 and low byte of LeftWheel in DID FD10 is B5.

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